Theory of Commutative Algebra 13

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Lecture notes with worked examples in commutative algebra

Contents

1 More Examples and Counterexamples: Spectra, Neighborhoods, Closed Sets, Ideals, Quotients, and Tensoring

A good way to deepen the subject is to proceed in the following order:

scheme $\longrightarrow$ spectrum $\longrightarrow$ points $\longrightarrow$ neighborhoods $\longrightarrow$ closed sets $\longrightarrow$ ideals and quotients $\longrightarrow$

At each stage one should look at:

- what the object means,
- standard examples,
- counterexamples or phenomena that differ from ordinary geometric intuition.

1.1 What is Spec(A)?

For a commutative ring A, one defines

$$\mathrm{Spec}(A) = \{\mathfrak{p} \subset A : \ \mathfrak{p} \text{ is a prime ideal}\}.$$

Thus a point of $\mathrm{Spec}(A)$ is a prime ideal.

This is the first major conceptual shift: in scheme theory, points are not initially numbers or coordinate tuples; they are prime ideals.

Example: $\mathrm{Spec}(k[t])$

Assume first that k is algebraically closed. Then the prime ideals of $k[t]$ are:

$$(0), \qquad (t-a) \text{ for } a \in k.$$

So $\mathrm{Spec}(k[t])$ consists of:

- one generic point (0),
- one closed point $(t-a)$ for each $a \in k$.

Example: $\mathrm{Spec}(\mathbb{Z})$

The prime ideals of $\mathbb{Z}$ are:

$$(0),\ (2),\ (3),\ (5), \ldots$$

Hence $\mathrm{Spec}(\mathbb{Z})$ consists of:

- one generic point (0),
- one closed point (p) for each prime number p.

Counterexample to naive geometry

Not every point of $\mathrm{Spec}(A)$ is closed.

For instance, in $\mathrm{Spec}(k[t])$, the point

$$(0)$$

is not closed. In fact,

$$\overline{\{(0)\}} = \mathrm{Spec}(k[t]).$$

So scheme-theoretic points are more general than ordinary geometric points.

1.2 Closed sets in $\mathrm{Spec}(A)$

For an ideal $I \subset A$, define

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

These sets are exactly the Zariski closed sets.

Example: one closed point in $k[t]$

Take

$$I = (t - a).$$

Then

$$V(t - a) = \{(t - a)\}.$$

So this is a single closed point.

Example: the whole space

Take

$$I = (0).$$

Then every prime ideal contains 0, so

$$V(0) = \mathrm{Spec}(k[t]).$$

Example: one point in the affine plane

In $k[s, t]$, take

$$I = (s - a, t - b).$$

Then

$$V(I)$$

is the single point (a, b).

Example: union of two lines

Now take

$$I = ((s-a)(t-b)).$$

Then

$$V(I) = V(s-a) \cup V(t-b).$$

So this is the union of the two lines

$$s = a \qquad \text{and} \qquad t = b.$$

Counterexample: different ideals can define the same closed set

In $k[x]$,

$$V(x) = V(x^2).$$

Indeed, a prime ideal contains x^2 if and only if it contains x.

So topologically these define the same closed set.

However, the quotient rings

$$k[x]/(x) \qquad \text{and} \qquad k[x]/(x^2)$$

are not the same.

This is one of the most important lessons:

$$\text{same closed set} \neq \text{same scheme structure.}$$

1.3 Open sets and neighborhoods

The basic open sets are

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) : f \notin \mathfrak{p}\}.$$

These form a basis for the Zariski topology.

A neighborhood of a point $x = \mathfrak{p}$ is any open set containing $\mathfrak{p}$. A basic neighborhood is some $D(f)$ with

$$f \notin \mathfrak{p}.$$

Example: neighborhoods of a closed point in $\operatorname{Spec}(k[t])$

Take the point

$$\mathfrak{p} = (t-a).$$

A basic open set $D(g)$ contains $(t-a)$ if and only if

$$g \notin (t-a),$$

which is equivalent to

$$g(a) \neq 0.$$

So the basic neighborhoods of $(t-a)$ are exactly the $D(g)$ where g does not vanish at a.

Example: neighborhoods of the generic point

Take the point

$$(0) \in \mathrm{Spec}(k[t]).$$

A basic open set $D(f)$ contains (0) if and only if

$$f \notin (0),$$

that is, if and only if $f \neq 0$.

Thus every nonempty basic open set contains the generic point.

So the generic point is dense in the whole space.

Counterexample to Hausdorff intuition

In ordinary topology, distinct points are often separable by disjoint neighborhoods. This usually fails in the Zariski topology.

For example, in $\mathrm{Spec}(k[t])$, the generic point (0) cannot be separated from any closed point by disjoint open sets.

So $\mathrm{Spec}(A)$ is generally not Hausdorff.

1.4 Closures of points

If $x = \mathfrak{p} \in \mathrm{Spec}(A)$, then

$$\overline{\{x\}} = V(\mathfrak{p}).$$

So the closure of a point is determined by the prime ideal itself.

Example: closed points

In $k[t]$, for the maximal ideal $(t-a)$,

$$\overline{\{(t-a)\}} = V(t-a) = \{(t-a)\}.$$

So maximal ideals correspond to closed points.

Example: the generic point

For the point (0),

$$\overline{\{(0)\}} = V(0) = \mathrm{Spec}(k[t]).$$

Example: generic points of components

Let

$$A = k[x,y]/(xy).$$

Then the ideals (x) and (y) are prime in A, and they are the generic points of the two irreducible components.

Their closures are:

$$\overline{\{(x)\}} = V(x), \qquad \overline{\{(y)\}} = V(y).$$

So a reducible scheme can have several generic points.

1.5 Local rings and residue fields

Let $x = \mathfrak{p} \in \operatorname{Spec}(A)$. Then the local ring at x is

$$\mathcal{O}_{X,x} = A_{\mathfrak{p}},$$

and the residue field is

$$k(x) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

Example: $k[t]$ **at** $(t-a)$

At the point $(t-a)$, the local ring is

$$k[t]_{(t-a)}.$$

This consists of fractions f/g with

$$g(a) \neq 0.$$

Its residue field is

$$k(x) \cong k.$$

Example: $k[t]$ **at** (0)

At the point (0),

$$\mathcal{O}_{X,(0)} = k(t),$$

because localizing a domain at the zero ideal gives its fraction field.

Thus

$$k((0)) = k(t).$$

Example: $\mathbb{R}[t]$ **at** (t^2+1)

Consider the prime ideal

$$(t^2+1) \subset \mathbb{R}[t].$$

Then

$$k(x) = \mathbb{R}[t]/(t^2+1) \cong \mathbb{C}.$$

So this is a closed point whose residue field is not $\mathbb{R}$.

Counterexample to the idea "closed point = base field"

A closed point of a scheme over k need not have residue field k.

It may have a larger field as residue field. For example, over $\mathbb{R}$, the closed point (t^2+1) has residue field $\mathbb{C}$.

1.6 Ideals and quotient rings

If $I \subseteq A$, then the quotient ring

$$A/I$$

defines the closed subscheme

$$\mathrm{Spec}(A/I) \subseteq \mathrm{Spec}(A).$$

This is how equations are imposed scheme-theoretically.

Example: one reduced point

In $k[s,t]$,

$$A = k[s,t]/(s-a, t-b) \cong k.$$

This is one reduced point.

Example: a doubled point

In $k[x]$,

$$A = k[x]/(x^2).$$

This is not the same as

$$k[x]/(x) \cong k.$$

Topologically both correspond to the same single point, but scheme-theoretically they differ because

$$\overline{x} \neq 0, \qquad \overline{x}^2 = 0.$$

So $k[x]/(x^2)$ has a nonzero nilpotent element.

This is a basic nonreduced scheme.

Example: two crossing lines

Consider

$$A = k[s,t]/(st).$$

This is the union of the two coordinate axes.

It has zero divisors:

$$\overline{s} \neq 0, \qquad \overline{t} \neq 0, \qquad \overline{s}\,\overline{t} = 0.$$

Counterexample: same topological space, different quotient ring

The rings

$$k[x]/(x) \qquad \text{and} \qquad k[x]/(x^2)$$

define the same underlying closed set $V(x)$, but not the same scheme.

So quotient rings retain more information than the set of zeros alone.

1.7 Reduced and nonreduced schemes

A ring is called reduced if it has no nonzero nilpotent elements.

Examples of reduced rings

The rings

$$k[x], \qquad k[s,t]/(st), \qquad \mathbb{Z}$$

are reduced.

Example of a nonreduced ring

The ring

$$k[x]/(x^2)$$

is not reduced.

Counterexample: one point can still carry extra structure

A nonreduced scheme may have only one point topologically but still contain extra infinitesimal information through nilpotent elements.

This is one of the main ways in which schemes are richer than classical varieties.

1.8 Irreducible and reducible spectra

Example: irreducible spectra

If A is a domain, then $\mathrm{Spec}(A)$ is irreducible.

So

$$\mathrm{Spec}(k[t]), \qquad \mathrm{Spec}(k[s,t]), \qquad \mathrm{Spec}(\mathbb{Z})$$

are irreducible.

Example: reducible spectrum

The ring

$$A = k[s,t]/(st)$$

is not a domain, and

$$\mathrm{Spec}(A) = V(s) \cup V(t)$$

is reducible.

Counterexample: reduced does not imply irreducible

The ring $k[s,t]/(st)$ is reduced but reducible.

So the properties

- reduced,
- irreducible

are not the same.

Reduced means no nilpotents. Irreducible means one irreducible topological component.

1.9 Tensor products and base change

Now we come to tensor products and base change.

If $k \to K$ is a field extension and A is a k-algebra, then base change is given algebraically by

$$A \longmapsto A \otimes_k K.$$

Geometrically, this corresponds to

$$\mathrm{Spec}(A) \longmapsto \mathrm{Spec}(A \otimes_k K).$$

Example: enlarging coefficients

One has

$$\mathbb{R}[t] \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[t].$$

So tensoring enlarges the coefficient field.

Quotients commute with tensoring

A fundamental fact is:

$$(A/I) \otimes_A B \cong B/IB.$$

In the polynomial setting, this becomes

$$\Big(k[x_1,\dots,x_n]/I\Big) \otimes_k K \cong K[x_1,\dots,x_n]/IK.$$

This is the key bridge between quotients and base change.

1.10 Step-by-step tensoring examples

Example: a linear factor

Let

$$A = \mathbb{R}[t]/(t-1) \cong \mathbb{R}.$$

Tensor with $\mathbb{C}$:

$$A \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[t]/(t-1) \cong \mathbb{C}.$$

Nothing splits.

Example: an irreducible quadratic

Let

$$A = \mathbb{R}[t]/(t^2+1) \cong \mathbb{C}.$$

Now tensor with $\mathbb{C}$:

$$A \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[t]/(t^2+1).$$

But

$$t^2+1 = (t-i)(t+i)$$

in $\mathbb{C}[t]$, so

$$\mathbb{C}[t]/(t^2+1) \cong \mathbb{C} \times \mathbb{C}.$$

Thus one real closed point with residue field $\mathbb{C}$ splits into two complex points.

Example: an equation already split

Let

$$A = \mathbb{R}[s,t]/(st).$$

Then

$$A \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[s,t]/(st).$$

No new splitting occurs because the equation was already factored over $\mathbb{R}$.

1.11 Counterexamples in tensoring

Counterexample: a prime ideal need not stay prime

In $\mathbb{R}[t]$, the ideal

$$(t^2+1)$$

is prime.

But after base change to $\mathbb{C}$, the corresponding ideal in $\mathbb{C}[t]$ is generated by

$$t^2+1 = (t-i)(t+i),$$

which is no longer prime.

So prime ideals can split after base change.

Counterexample: irreducible over k need not remain irreducible over K

The polynomial

$$t^2 + 1$$

is irreducible over $\mathbb{R}$, but reducible over $\mathbb{C}$.

Counterexample: same object over one field, different geometry over another

Over $\mathbb{R}$,

$$\mathbb{R}[t]/(t^2 + 1)$$

corresponds to one closed point.

After base change to $\mathbb{C}$, it becomes

$$\mathbb{C}[t]/(t^2 + 1) \cong \mathbb{C} \times \mathbb{C},$$

so geometrically it becomes two points.

Thus the geometry depends strongly on the base field.

1.12 Comparison table

It is useful to compare several standard quotient rings.

One reduced point

$$k[s, t]/(s - a, t - b) \cong k.$$

Interpretation: one reduced point.

One doubled point

$$k[x]/(x^2).$$

Interpretation: one nonreduced point.

Two reduced points

$$k[t]/((t - a)(t - b)) \cong k \times k \qquad (a \neq b).$$

Interpretation: two reduced points.

Two lines crossing

$$k[s, t]/(st).$$

Interpretation: two irreducible components crossing.

One non-rational real point

$$\mathbb{R}[t]/(t^2+1) \cong \mathbb{C}.$$

Interpretation: one closed point over $\mathbb{R}$ with residue field $\mathbb{C}$.

After complexification

$$\Big(\mathbb{R}[t]/(t^2+1)\Big) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}.$$

Interpretation: the real point splits into two complex points.

1.13 Good practice exercises

The following are very useful exercises for building intuition.

1. Describe all points of
$$\operatorname{Spec}(k[t]).$$
2. Compute the residue fields of
$$(0), \qquad (t-a), \qquad (t^2+1) \subset \mathbb{R}[t].$$
3. Compare
$$k[x]/(x) \qquad \text{and} \qquad k[x]/(x^2).$$
Do they have the same topological space? Do they define the same scheme?
4. Describe the closed sets of
$$\operatorname{Spec}(\mathbb{Z}).$$
5. Show that
$$V(xy) = V(x) \cup V(y)$$
in $k[x,y]$.
6. Compute
$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}.$$
7. Show that
$$\Big(k[x]/(f)\Big) \otimes_k K \cong K[x]/(f).$$

1.14 Final summary

The main point of all these examples and counterexamples is that scheme theory keeps track simultaneously of:

- the topological space $\operatorname{Spec}(A)$,
- the algebra of functions on it,
- local behavior through local rings and residue fields,

- scheme structure through quotients,
- and behavior under change of base field through tensor products.

The most important warnings are:

- not every point is closed,
- the Zariski topology is very far from Hausdorff,
- different ideals can define the same closed set,
- the same topological space can carry different scheme structures,
- reduced and irreducible are different notions,
- prime ideals may split after base change,
- geometry can change dramatically when the base field changes.

These are precisely the phenomena that make scheme theory richer and more powerful than ordinary affine geometry.

1.15 Can a prime ideal contain another ideal?

Yes. A prime ideal is itself an ideal, so it can contain other ideals.

If A is a ring and

$$\mathfrak{p} \subset A$$

is a prime ideal, then for many ideals $I \subset A$ one may have

$$I \subseteq \mathfrak{p}.$$

This is in fact one of the central ideas in the definition of closed sets in $\mathrm{Spec}(A)$.

Why this matters in $\mathrm{Spec}(A)$

Recall that for an ideal $I \subseteq A$, the corresponding Zariski closed set is

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

Thus a point $\mathfrak{p} \in \mathrm{Spec}(A)$ lies in the closed set $V(I)$ precisely when the prime ideal $\mathfrak{p}$ contains the ideal I.

So the geometry of $\mathrm{Spec}(A)$ is built from the containment relation between ideals.

What ideal containment means

The statement

$$I \subseteq \mathfrak{p}$$

means that every element of the ideal I belongs to the ideal $\mathfrak{p}$.

So this is *ideal containment*, that is, containment of subsets of the ring A. It is not merely a statement about one element. It means:

$$\forall f \in I, \quad f \in \mathfrak{p}.$$

Since an ideal is a subset of the ring, one ideal may be contained in another.

Example in $k[t]$

Let
$$A = k[t].$$
Take the ideal
$$I = (t^2)$$
and the prime ideal
$$\mathfrak{p} = (t).$$

Then
$$(t^2) \subseteq (t).$$

Indeed, every element of (t^2) has the form
$$t^2 g(t)$$
for some $g(t) \in k[t]$. But such an element is certainly divisible by t, hence belongs to (t).

Therefore the prime ideal (t) contains the ideal (t^2).

Consequently,
$$(t) \in V(t^2).$$

Example in $k[s,t]$

Let
$$A = k[s,t].$$
Take the ideal
$$I = ((s-a)(t-b))$$
and the prime ideal
$$\mathfrak{p} = (s-a, t-b).$$

Then
$$((s-a)(t-b)) \subseteq (s-a, t-b).$$

This is because the generator
$$(s-a)(t-b)$$
belongs to the ideal $(s-a, t-b)$: it is a multiple of $s-a$, and any multiple of $s-a$ lies in $(s-a, t-b)$.

Since the ideal $((s-a)(t-b))$ consists of all multiples of $(s-a)(t-b)$, every element of it also lies in $(s-a, t-b)$.

So the point $(s-a, t-b)$ lies in
$$V((s-a)(t-b)).$$

Geometrically, this means that the point (a, b) lies on the closed set defined by

$$(s-a)(t-b)=0,$$

that is, on the union of the two lines

$$s=a \qquad \text{and} \qquad t=b.$$

Example in $\mathbb{Z}$

Let

$$A=\mathbb{Z}.$$

Take

$$I=(6)$$

and the prime ideal

$$\mathfrak{p}=(2).$$

Then

$$(6)\subseteq(2),$$

because every multiple of 6 is also a multiple of 2.

Similarly,

$$(6)\subseteq(3).$$

Thus the prime ideals containing (6) are exactly

$$(2) \qquad \text{and} \qquad (3).$$

So in $\mathrm{Spec}(\mathbb{Z})$,

$$V(6)=\{(2),(3)\}.$$

This is a very important arithmetic example.

Another example in $k[x, y]$

Take

$$A=k[x,y], \qquad I=(xy), \qquad \mathfrak{p}=(x).$$

Then

$$(xy)\subseteq(x),$$

because xy is a multiple of x.

Likewise,

$$(xy)\subseteq(y).$$

Hence both prime ideals (x) and (y) lie in

$$V(xy).$$

This matches the geometric identity

$$V(xy)=V(x)\cup V(y),$$

which says that the zero set of xy is the union of the two coordinate axes.

A useful general observation

If

$$I = (f)$$

is a principal ideal and $f \in \mathfrak{p}$, then automatically

$$(f) \subseteq \mathfrak{p}.$$

Indeed, every element of (f) has the form

$$fg$$

for some $g \in A$, and since $\mathfrak{p}$ is an ideal and $f \in \mathfrak{p}$, we have

$$fg \in \mathfrak{p}.$$

So for principal ideals, to check

$$(f) \subseteq \mathfrak{p}$$

it is enough to check that the generator f lies in $\mathfrak{p}$.

More generally, if

$$I = (f_1, \ldots, f_r),$$

then

$$I \subseteq \mathfrak{p}$$

if and only if

$$f_1, \ldots, f_r \in \mathfrak{p}.$$

Connection with closed sets

This containment relation is exactly what makes the notation

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec}(A) : I \subseteq \mathfrak{p}\}$$

natural.

A prime ideal $\mathfrak{p}$ is a point of $\operatorname{Spec}(A)$, and it lies in $V(I)$ precisely when all equations in the ideal I vanish at that point in the scheme-theoretic sense.

Thus closed sets in $\operatorname{Spec}(A)$ are determined by asking which prime ideals contain a given ideal.

Final summary

So the answer is yes:

- a prime ideal is itself an ideal,
- therefore it can contain other ideals,
- and this containment is one of the basic ideas used to define closed sets in $\operatorname{Spec}(A)$.

In symbols, if $I \subseteq A$ and $\mathfrak{p} \in \mathrm{Spec}(A)$, then

$$\mathfrak{p} \in V(I) \qquad \Longleftrightarrow \qquad I \subseteq \mathfrak{p}.$$

Thus prime ideals very often contain other ideals, and the relation

$$I \subseteq \mathfrak{p}$$

is one of the central relations in scheme theory.

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1.16 What is the ideal (0) in $k[x]$?

In the polynomial ring

$$k[x],$$

the notation

$$(0)$$

means the *zero ideal.* This is the ideal consisting only of the zero polynomial:

$$(0) = \{0\}.$$

So (0) is not the same as (x), $(x - a)$, or any other nonzero principal ideal. It contains only one element, namely 0.

Why (0) is an ideal

It is an ideal because:

- it is closed under addition, since

$$0 + 0 = 0,$$

- it is closed under multiplication by arbitrary elements of $k[x]$, since for every $f(x) \in k[x]$,

$$f(x) \cdot 0 = 0.$$

Thus (0) satisfies the definition of an ideal.

Why (0) is prime in $k[x]$

Since k is a field, the ring $k[x]$ is an integral domain. That means if

$$f(x)g(x) = 0$$

in $k[x]$, then necessarily

$$f(x) = 0 \qquad \text{or} \qquad g(x) = 0.$$

Now an ideal $I \subseteq A$ is prime if whenever

$$ab \in I,$$

then

$$a \in I \qquad \text{or} \qquad b \in I.$$

For the zero ideal (0), this means:

if

$$f(x)g(x) \in (0),$$

then

$$f(x)g(x) = 0,$$

and since $k[x]$ is a domain, this implies

$$f(x) = 0 \qquad \text{or} \qquad g(x) = 0,$$

that is,

$$f(x) \in (0) \qquad \text{or} \qquad g(x) \in (0).$$

Therefore (0) is a prime ideal in $k[x]$.

Why (0) is not maximal

An ideal I is maximal if the quotient ring A/I is a field.

Now

$$k[x]/(0) \cong k[x].$$

But $k[x]$ is not a field, because for example x is not invertible in $k[x]$.

So (0) is not maximal.

Thus in $k[x]$, the ideal (0) is:

- an ideal,
- a prime ideal,
- not a maximal ideal.

What point does (0) give in $\mathrm{Spec}(k[x])$?

Since points of

$$\mathrm{Spec}(k[x])$$

are prime ideals, the ideal (0) defines a point of $\mathrm{Spec}(k[x])$.

This point is called the *generic point* of the affine line.

The reason is that its closure is the whole space:

$$\overline{\{(0)\}} = V((0)).$$

But every prime ideal contains 0, so

$$V((0)) = \mathrm{Spec}(k[x]).$$

Thus the point (0) is dense in the whole spectrum.

Geometric meaning

Geometrically, the ideal (0) does not correspond to one ordinary closed point such as $x = a$. Instead, it corresponds to the whole irreducible affine line viewed generically.

By contrast:

- $(x - a)$ is a closed point,
- (0) is the generic point of the whole space.

Comparison with (x)

It is useful to compare:

$$(0) \qquad \text{and} \qquad (x).$$

We have:

$$(0) = \{0\},$$

whereas

$$(x) = \{xf(x) : f(x) \in k[x]\}.$$

So (x) contains many polynomials, namely all multiples of x, while (0) contains only the zero polynomial.

Also,

$$(0) \subsetneq (x).$$

Final summary

In the ring

$$k[x],$$

the symbol

$$(0)$$

means the zero ideal:

$$(0) = \{0\}.$$

It is prime because $k[x]$ is an integral domain, but it is not maximal because

$$k[x]/(0) \cong k[x]$$

is not a field.

Therefore (0) is a point of

$$\mathrm{Spec}(k[x]),$$

and this point is the generic point of the affine line.

2 Localization, Distinguished Opens $D(f)$, Stalks, and Local Computations

The next natural step after spectra, closed sets, ideals, quotients, and residue fields is to understand how one passes from *global* algebra to *local* algebra.

The guiding chain is:

$$A \longrightarrow D(f) \longrightarrow A_f \longrightarrow A_{\mathfrak{p}} \longrightarrow k(\mathfrak{p}).$$

Here:

- A is the global coordinate ring of an affine scheme $\operatorname{Spec}(A)$,
- $D(f)$ is a basic open subset,
- A_f is the ring of functions on that open subset,
- $A_{\mathfrak{p}}$ is the stalk, that is, the ring of functions defined near the point $\mathfrak{p}$,
- $k(\mathfrak{p})$ is the residue field, obtained by evaluating local functions at the point.

So localization is the algebraic mechanism for "zooming in."

2.1 Distinguished open sets $D(f)$

Let A be a ring and let $f \in A$.

Definition 2.1. The *distinguished open set* determined by f is

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) : f \notin \mathfrak{p}\}.$$

Equivalently,

$$D(f) = \operatorname{Spec}(A) \setminus V(f),$$

where

$$V(f) = V((f)) = \{\mathfrak{p} \in \operatorname{Spec}(A) : f \in \mathfrak{p}\}.$$

Thus $D(f)$ is the open set where f does not vanish in the scheme-theoretic sense.

First intuition

The open set $D(f)$ should be thought of as the region where f is allowed to be invertible.

This is exactly what will happen algebraically after localization.

Example: $D(t)$ in $\operatorname{Spec}(k[t])$

Let

$$A = k[t].$$

Then

$$D(t) = \{\mathfrak{p} \in \operatorname{Spec}(k[t]) : t \notin \mathfrak{p}\}.$$

If k is algebraically closed, the points of $\mathrm{Spec}(k[t])$ are:

$$(0), \qquad (t-a) \text{ for } a \in k.$$

Now:

- the prime (t) is not in $D(t)$, because $t \in (t)$,
- a prime $(t-a)$ with $a \neq 0$ is in $D(t)$, because $t \notin (t-a)$,
- the generic point (0) is also in $D(t)$, because $t \neq 0$, so $t \notin (0)$.

So $D(t)$ is the affine line with the closed point $t = 0$ removed, but the generic point remains.

Example: $D(s)$ in $\mathrm{Spec}(k[s,t])$

Let

$$A = k[s,t].$$

Then

$$D(s) = \mathrm{Spec}(k[s,t]) \setminus V(s).$$

Geometrically this is the complement of the line

$$s = 0.$$

So $D(s)$ is the open set where the coordinate s does not vanish.

Counterexample to naive pointwise interpretation

One should not interpret $D(f)$ too naively as "the set of coordinate points where $f \neq 0$," because points of $\mathrm{Spec}(A)$ are prime ideals, not just tuples of coordinates.

The correct statement is:

$$\mathfrak{p} \in D(f) \qquad \Longleftrightarrow \qquad f \notin \mathfrak{p}.$$

2.2 Distinguished opens form a basis

The sets $D(f)$ form a basis for the Zariski topology on $\mathrm{Spec}(A)$.

This means that every open subset of $\mathrm{Spec}(A)$ is a union of distinguished opens.

So to understand neighborhoods in $\mathrm{Spec}(A)$, it is enough to understand the sets $D(f)$.

Neighborhoods of a point

Let $x = \mathfrak{p} \in \mathrm{Spec}(A)$. Then a distinguished open $D(f)$ contains x if and only if

$$f \notin \mathfrak{p}.$$

Thus the basic neighborhoods of a point are determined by those elements of A that do not lie in the corresponding prime ideal.

Example: neighborhoods of $(t-a)$ in $\mathrm{Spec}(k[t])$

Let
$$\mathfrak{p} = (t-a).$$
Then $D(g)$ contains $(t-a)$ if and only if
$$g \notin (t-a),$$
which is equivalent to
$$g(a) \neq 0.$$
So the basic neighborhoods of $(t-a)$ are precisely the distinguished opens $D(g)$ where g does not vanish at a.

Example: neighborhoods of the generic point

Let
$$\mathfrak{p} = (0) \in \mathrm{Spec}(k[t]).$$
Then $D(f)$ contains (0) if and only if
$$f \notin (0),$$
that is, if and only if $f \neq 0$.

Thus every nonempty distinguished open contains the generic point.

So the generic point is dense.

Counterexample to Hausdorff intuition

In ordinary topology, one often expects two different points to have disjoint neighborhoods. This usually fails in the Zariski topology.

For example, in $\mathrm{Spec}(k[t])$, the generic point (0) cannot be separated from any closed point by disjoint open sets.

So affine schemes are generally not Hausdorff spaces.

2.3 Localization A_f

Let A be a ring and $f \in A$.

Definition 2.2. The *localization of A at f* is
$$A_f = S^{-1}A, \qquad S = \{1, f, f^2, f^3, \dots\}.$$

So A_f is obtained by formally inverting f.

An element of A_f is represented by a fraction
$$\frac{a}{f^n}, \qquad a \in A,\ n \geq 0.$$

Geometric meaning

The ring A_f is the ring of regular functions on the open set $D(f)$:

$$\Gamma(D(f), \mathcal{O}_{\mathrm{Spec}(A)}) = A_f.$$

Thus localization at f corresponds to restricting from the whole affine scheme $\mathrm{Spec}(A)$ to the open subset $D(f)$.

Example: $k[t]$ localized at t

Let

$$A = k[t].$$

Then

$$A_t = k[t, t^{-1}].$$

So on the open set $D(t)$, one may use functions such as

$$\frac{1}{t}, \qquad \frac{t^2+1}{t^3}, \qquad t + \frac{1}{t}.$$

These are regular on $D(t)$, but not globally on all of $\mathrm{Spec}(k[t])$.

Example: $k[s, t]$ localized at s

Let

$$A = k[s, t].$$

Then

$$A_s = k[s, t, 1/s].$$

So on $D(s)$, one may use expressions involving powers of $1/s$, for example

$$\frac{t}{s}, \qquad \frac{s^2+t}{s^3}.$$

Counterexample: local regularity does not imply global regularity

The function

$$\frac{1}{t}$$

belongs to

$$k[t]_t,$$

but it does not belong to $k[t]$.

So a function may be regular on an open set without being globally regular.

This is one of the main reasons the structure sheaf is essential.

2.4 The distinguished open $D(f)$ is affine

One of the most important facts about affine schemes is that distinguished opens are again affine.

Theorem 2.3. Let A be a ring and $f \in A$. Then

$$D(f) \cong \operatorname{Spec}(A_f).$$

So every distinguished open set is itself an affine scheme.

Example: punctured affine line as a distinguished open

In $\operatorname{Spec}(k[t])$,

$$D(t) \cong \operatorname{Spec}(k[t, t^{-1}]).$$

So the affine line minus the point $t = 0$ is still affine.

Counterexample: not every open subset of an affine scheme is affine

Although distinguished opens are affine, an arbitrary open subset of an affine scheme need not be affine.

So one must not confuse the statement

$$D(f) \text{ is affine}$$

with the false statement

$$\text{every open subset of an affine scheme is affine.}$$

2.5 Localization at a prime ideal

Let $\mathfrak{p} \in \operatorname{Spec}(A)$.

Definition 2.4. The *localization of A at the prime ideal* $\mathfrak{p}$ is

$$A_{\mathfrak{p}} = S^{-1}A, \qquad S = A \setminus \mathfrak{p}.$$

So in $A_{\mathfrak{p}}$, one inverts every element of A not belonging to $\mathfrak{p}$.

Geometric meaning

The ring $A_{\mathfrak{p}}$ is the stalk of the structure sheaf at the point $\mathfrak{p}$:

$$\mathcal{O}_{\operatorname{Spec}(A),\mathfrak{p}} = A_{\mathfrak{p}}.$$

This ring should be thought of as the ring of functions defined near the point $\mathfrak{p}$.

So A_f is "functions on one chosen open set," whereas $A_\mathfrak{p}$ is "functions defined in some neighborhood of the point."

Example: $k[t]$ at the closed point $(t-a)$

Let

$$\mathfrak{p} = (t-a).$$

Then

$$k[t]_\mathfrak{p} = k[t]_{(t-a)}$$

consists of fractions

$$\frac{g(t)}{h(t)}$$

such that

$$h(a) \neq 0.$$

So one may divide by any polynomial that does not vanish at the point a.

Example: $k[t]$ at the generic point

Let

$$\mathfrak{p} = (0).$$

Then one inverts every nonzero polynomial, so

$$k[t]_{(0)} = k(t),$$

the field of fractions.

Thus the generic point sees rational functions.

Example: $\mathbb{Z}$ at (p)

Let

$$\mathfrak{p} = (p) \subset \mathbb{Z}.$$

Then

$$\mathbb{Z}_{(p)}$$

consists of fractions

$$\frac{a}{b}$$

with

$$p \nmid b.$$

So the only forbidden denominators are those divisible by p.

2.6 The relation between A_f and $A_\mathfrak{p}$

If

$$\mathfrak{p} \in D(f),$$

then

$$f \notin \mathfrak{p}.$$

So f becomes invertible in $A_\mathfrak{p}$.

Therefore there is a natural ring map

$$A_f \longrightarrow A_\mathfrak{p}.$$

Geometrically, this means that a function regular on the open set $D(f)$ determines a germ of a regular function at every point of $D(f)$.

Interpretation

Thus

$$A_f$$

describes regular functions on a specific neighborhood, while

$$A_\mathfrak{p}$$

collects the behavior of regular functions on arbitrarily small neighborhoods of the point $\mathfrak{p}$.

2.7 The stalk as a direct limit

A useful conceptual description is:

$$A_\mathfrak{p} \cong \varinjlim_{\mathfrak{p} \in D(f)} A_f.$$

This means that the stalk is obtained by taking all distinguished neighborhoods of the point $\mathfrak{p}$, looking at their rings of sections, and identifying sections that agree on smaller neighborhoods.

Even if one does not use direct limits formally at first, the idea is important:

> The stalk is the algebra of germs of functions near the point.

2.8 The local ring is a local ring

The ring $A_\mathfrak{p}$ is called a *local ring* because it has a unique maximal ideal.

Proposition 2.5. The unique maximal ideal of $A_\mathfrak{p}$ is

$$\mathfrak{p}A_\mathfrak{p}.$$

The nonunits of $A_\mathfrak{p}$ are exactly the elements of $\mathfrak{p}A_\mathfrak{p}$.

Example: $k[t]_{(t-a)}$

In

$$k[t]_{(t-a)},$$

the unique maximal ideal is generated by $t - a$.

A fraction

$$\frac{g}{h}$$

is invertible if and only if

$$g(a) \neq 0.$$

So the only obstruction to invertibility is vanishing at the point a.

Example: $k[s,t]_{(s-a,t-b)}$

In

$$k[s,t]_{(s-a,t-b)},$$

one may divide by any polynomial $g(s,t)$ such that

$$g(a,b) \neq 0.$$

But one cannot divide by

$$s - a \qquad \text{or} \qquad t - b,$$

because these lie in the maximal ideal.

2.9 Residue fields from stalks

Once one has the stalk $A_{\mathfrak{p}}$, the residue field is obtained by quotienting by its maximal ideal:

$$k(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

This is the field of values of local functions at the point $\mathfrak{p}$.

Example: $k[t]$ **at** $(t-a)$

One has

$$k[t]_{(t-a)}/(t-a)k[t]_{(t-a)} \cong k.$$

Example: $k[t]$ **at** (0)

One has

$$k[t]_{(0)}/(0) \cong k(t).$$

So the generic point has residue field the rational function field.

Example: $\mathbb{R}[t]$ **at** (t^2+1)

The residue field is

$$\mathbb{R}[t]_{(t^2+1)}/(t^2+1)\mathbb{R}[t]_{(t^2+1)} \cong \mathbb{R}[t]/(t^2+1) \cong \mathbb{C}.$$

So this point has residue field $\mathbb{C}$, even though the scheme is defined over $\mathbb{R}$.

2.10 Local computations on distinguished opens

The ring A_f allows concrete calculations on the open set $D(f)$.

Example: functions on $D(t) \subset \operatorname{Spec}(k[t])$

On the open set $D(t)$, the ring of regular functions is

$$k[t]_t = k[t, t^{-1}].$$

Thus functions such as

$$\frac{1}{t}, \qquad \frac{t^2+1}{t^3}, \qquad t + \frac{1}{t}$$

are regular on $D(t)$.

But

$$\frac{1}{t}$$

is not regular on all of $\operatorname{Spec}(k[t])$, because it is not defined at the point (t).

Example: functions on $D(s) \subset \operatorname{Spec}(k[s,t])$

On the open set $D(s)$, the ring of regular functions is

$$k[s,t]_s.$$

Thus expressions such as

$$\frac{t}{s}, \qquad \frac{s^2+t}{s^3}$$

are regular on $D(s)$.

Counterexample: regularity is local

The function

$$\frac{1}{s}$$

is regular on $D(s)$, but not on $D(t)$, and not globally on $\operatorname{Spec}(k[s,t])$.

So regularity depends on the open set under consideration.

2.11 Local computations at a point

Now one can compute what is allowed near a specific point.

Example: $k[t]$ **at** $(t-a)$

In

$$k[t]_{(t-a)},$$

the fraction

$$\frac{1}{t-a}$$

does not exist, because $t-a \in (t-a)$, so it is not inverted.
But

$$\frac{1}{t-b}$$

does exist if $b \neq a$, because then

$$t-b \notin (t-a).$$

So one may divide by functions that do not vanish at the point.

Example: $k[s,t]$ **at** $(s-a, t-b)$

In

$$k[s,t]_{(s-a,t-b)},$$

one may divide by any polynomial $g(s,t)$ with

$$g(a,b) \neq 0.$$

But one cannot divide by

$$s-a \qquad \text{or} \qquad t-b.$$

This is exactly the local picture of regularity near a point.

2.12 Important identities for distinguished opens

The distinguished opens satisfy several useful identities.

Proposition 2.6. For any ring A and $f, g \in A$,

$$D(fg) = D(f) \cap D(g).$$

Also,

$$D(1) = \operatorname{Spec}(A), \qquad D(0) = \varnothing.$$

Proof. A prime ideal $\mathfrak{p}$ lies in $D(fg)$ if and only if

$$fg \notin \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$f \notin \mathfrak{p} \quad \text{and} \quad g \notin \mathfrak{p},$$

that is,

$$\mathfrak{p} \in D(f) \cap D(g).$$

So

$$D(fg) = D(f) \cap D(g).$$

Also, 1 belongs to no proper ideal, so

$$D(1) = \mathrm{Spec}(A).$$

On the other hand, 0 belongs to every prime ideal, so

$$D(0) = \varnothing.$$

□

Counterexample: $D(f) = D(g)$ does not imply $f = g$

For example,

$$D(f) = D(f^2)$$

for every $f \in A$, because a prime ideal contains f if and only if it contains f^2.

So different elements can define the same distinguished open.

Counterexample: nilpotents give empty distinguished opens

If f is nilpotent, then f belongs to every prime ideal. Hence

$$D(f) = \varnothing.$$

For example, in

$$A = k[x]/(x^2),$$

the class $\overline{x}$ is nilpotent, so

$$D(\overline{x}) = \varnothing.$$

This is a purely scheme-theoretic phenomenon.

2.13 A nonreduced example: $k[x]/(x^2)$

Let

$$A = k[x]/(x^2).$$

This ring has only one prime ideal, namely

$$(\overline{x}).$$

So $\mathrm{Spec}(A)$ has only one point.

Since A is already local, the stalk at that point is the whole ring:

$$A_{(\overline{x})} = A.$$

Its maximal ideal is

$$(\overline{x}),$$

and its residue field is

$$A/(\overline{x}) \cong k.$$

But the ring is not reduced, because

$$\overline{x} \neq 0, \qquad \overline{x}^2 = 0.$$

So this is a one-point scheme carrying nontrivial infinitesimal structure.

Counterexample to classical intuition

Classically, one point should have no room for extra geometry. But in scheme theory, a one-point space can still have nontrivial algebra through nilpotent elements.

2.14 A reducible example: $k[s,t]/(st)$

Let

$$A = k[s,t]/(st).$$

Then $\mathrm{Spec}(A)$ is the union of the two coordinate axes.

At the crossing point

At the point corresponding to

$$(s,t),$$

the local ring is

$$A_{(s,t)}.$$

This ring contains zero divisors:

$$\overline{s} \neq 0, \qquad \overline{t} \neq 0, \qquad \overline{s}\,\overline{t} = 0.$$

So the singular crossing is visible already in the local ring.

At a smooth point of one component

If one localizes at a point on the s-axis away from the origin, then locally the other component disappears, because one of the coordinates becomes invertible.

So localizing at different points reveals different local geometry.

2.15 Good examples to practice

The following are particularly useful for computation and intuition:

1. $\mathrm{Spec}(k[t])$: distinguished opens, generic point, local rings at (0) and $(t-a)$.
2. $\mathrm{Spec}(k[s,t])$: opens $D(s)$, $D(t)$, and local rings at $(s-a, t-b)$.
3. $\mathrm{Spec}(k[x]/(x^2))$: nonreduced one-point scheme.
4. $\mathrm{Spec}(k[s,t]/(st))$: reducible scheme and local behavior at the crossing.
5. $\mathrm{Spec}(\mathbb{Z})$: local ring $\mathbb{Z}_{(p)}$ and generic stalk $\mathbb{Q}$.

2.16 Practice exercises

The following exercises are a natural next step.

1. Describe $D(t) \subset \mathrm{Spec}(k[t])$ and compute its ring of sections.
2. Show that
$$D(f) \cong \mathrm{Spec}(A_f).$$
3. Compute
$$k[t]_{(t-a)}$$
and describe its units.
4. Compute the local ring of
$$k[s,t]/(st)$$
at the point (s,t), and compare it with local rings at points on one component away from the intersection.
5. In
$$A = k[x]/(x^2),$$
compute $D(\overline{x})$.
6. Prove that
$$D(fg) = D(f) \cap D(g).$$
7. Prove that
$$D(1) = \mathrm{Spec}(A), \qquad D(0) = \varnothing.$$
8. Show that
$$D(f) = D(f^2).$$

2.17 Final summary

Localization is the algebraic language of local geometry.

The main picture to remember is:

- A describes global functions on $\mathrm{Spec}(A)$,
- $D(f)$ is the open region where f does not vanish,
- A_f is the ring of functions on that region,
- $A_\mathfrak{p}$ is the ring of functions defined near the point $\mathfrak{p}$,
- $k(\mathfrak{p})$ is the field of values of local functions at that point.

So the passage

$$A \longrightarrow A_f \longrightarrow A_\mathfrak{p} \longrightarrow k(\mathfrak{p})$$

is the algebraic form of passing from global geometry to local geometry and finally to the value field at a point.

The main warnings are:

- not every open set is affine, even though every $D(f)$ is affine,
- regularity is local, not purely global,
- different elements can define the same open set,
- nilpotents can make distinguished opens empty,
- one-point schemes can still carry nontrivial infinitesimal structure.

This is precisely why localization and stalks are central in the theory of schemes.

2.18 A collection of step-by-step examples for $A \to D(f) \to A_f \to A_\mathfrak{p} \to k$

The chain

$$A \longrightarrow D(f) \longrightarrow A_f \longrightarrow A_\mathfrak{p} \longrightarrow k(\mathfrak{p})$$

is one of the best ways to understand how algebra passes from global information to local information.

The meaning of each term is:

- A: the global coordinate ring,
- $D(f)$: the distinguished open set where $f \notin \mathfrak{p}$,
- A_f: the ring of regular functions on $D(f)$,
- $A_\mathfrak{p}$: the local ring at the point $\mathfrak{p}$,
- $k(\mathfrak{p})$: the residue field at that point.

The following examples illustrate this chain in several important rings.

Example 1: $A = k[x]$, $f = x$, $\mathfrak{p} = (x - a)$ with $a \neq 0$

Let

$$A = k[x].$$

We choose

$$f = x, \qquad \mathfrak{p} = (x - a), \qquad a \neq 0.$$

Since $a \neq 0$, we have

$$x \notin (x - a),$$

so

$$\mathfrak{p} \in D(x).$$

Now the chain is:

Global ring.

$$A = k[x].$$

Distinguished open.

$$D(x) = \{\mathfrak{q} \in \operatorname{Spec}(k[x]) : x \notin \mathfrak{q}\}.$$

Geometrically, this is the affine line with the closed point $x = 0$ removed.

Localization on the open.

$$A_x = k[x]_x = k[x, x^{-1}].$$

So on $D(x)$, the function $1/x$ is regular.

Local ring at the point.

$$A_{\mathfrak{p}} = k[x]_{(x-a)}.$$

This consists of fractions

$$\frac{g(x)}{h(x)}$$

such that

$$h(a) \neq 0.$$

Residue field.

$$k(\mathfrak{p}) = k[x]_{(x-a)}/(x - a)k[x]_{(x-a)} \cong k.$$

Thus the whole chain becomes

$$k[x] \longrightarrow D(x) \longrightarrow k[x, x^{-1}] \longrightarrow k[x]_{(x-a)} \longrightarrow k.$$

Example 2: $A = k[x]$**,** $f = x$**,** $\mathfrak{p} = (0)$

This is the generic-point version of the previous example.

Take

$$A = k[x], \qquad f = x, \qquad \mathfrak{p} = (0).$$

Since $x \neq 0$, we have

$$x \notin (0),$$

so

$$(0) \in D(x).$$

Now:

Global ring.

$$A = k[x].$$

Distinguished open.

$$D(x) = \operatorname{Spec}(k[x]) \setminus V(x).$$

Localization on the open.

$$A_x = k[x, x^{-1}].$$

Local ring at the generic point. Since $k[x]$ is a domain,

$$A_{(0)} = k(x),$$

the field of rational functions.

Residue field. The maximal ideal of $A_{(0)}$ is (0), so

$$k((0)) = k(x).$$

Thus the chain is

$$k[x] \longrightarrow D(x) \longrightarrow k[x, x^{-1}] \longrightarrow k(x) \longrightarrow k(x).$$

This shows that at the generic point, the local ring is already a field.

Example 3: $A = k[x, y]$**,** $f = x$**,** $\mathfrak{p} = (x - 1, y)$

Now consider the affine plane.

Let

$$A = k[x, y], \qquad f = x, \qquad \mathfrak{p} = (x - 1, y).$$

Since $x \notin (x - 1, y)$, the point $\mathfrak{p}$ lies in $D(x)$.

Global ring.

$$A = k[x, y].$$

Distinguished open.

$$D(x) = \mathrm{Spec}(k[x, y]) \setminus V(x).$$

Geometrically, this is the complement of the line $x = 0$.

Localization on the open.

$$A_x = k[x, y, 1/x].$$

Local ring at the point.

$$A_{\mathfrak{p}} = k[x, y]_{(x-1,y)}.$$

Residue field.

$$k(\mathfrak{p}) = k[x, y]_{(x-1,y)}/(x-1, y)k[x, y]_{(x-1,y)} \cong k.$$

Thus:

$$k[x, y] \longrightarrow D(x) \longrightarrow k[x, y, 1/x] \longrightarrow k[x, y]_{(x-1,y)} \longrightarrow k.$$

Example 4: $A = \mathbb{Z}$, $f = 2$, $\mathfrak{p} = (3)$

This is the arithmetic analogue of the polynomial examples.

Let

$$A = \mathbb{Z}, \qquad f = 2, \qquad \mathfrak{p} = (3).$$

Since

$$2 \notin (3),$$

we have

$$(3) \in D(2).$$

Global ring.

$$A = \mathbb{Z}.$$

Distinguished open.

$$D(2) = \{\mathfrak{q} \in \mathrm{Spec}(\mathbb{Z}) : 2 \notin \mathfrak{q}\}.$$

This removes the prime (2), but keeps (3) and the generic point (0).

Localization on the open.

$$A_2 = \mathbb{Z}[1/2].$$

Local ring at the point.

$$A_{(3)} = \mathbb{Z}_{(3)}.$$

Its elements are fractions

$$\frac{a}{b}$$

such that

$$3 \nmid b.$$

Residue field.

$$k((3)) = \mathbb{Z}_{(3)}/3\mathbb{Z}_{(3)} \cong \mathbb{F}_3.$$

Thus:

$$\mathbb{Z} \longrightarrow D(2) \longrightarrow \mathbb{Z}[1/2] \longrightarrow \mathbb{Z}_{(3)} \longrightarrow \mathbb{F}_3.$$

Example 5: $A = \mathbb{Z}$, $f = 6$, $\mathfrak{p} = (0)$

Now take the generic arithmetic point.

Let

$$A = \mathbb{Z}, \qquad f = 6, \qquad \mathfrak{p} = (0).$$

Since $6 \neq 0$, we have

$$6 \notin (0),$$

so

$$(0) \in D(6).$$

Global ring.

$$A = \mathbb{Z}.$$

Distinguished open.

$$D(6)$$

removes the points (2) and (3), but contains all other prime ideals and the generic point.

Localization on the open.

$$A_6 = \mathbb{Z}[1/6].$$

Local ring at the generic point.

$$A_{(0)} = \mathbb{Q}.$$

Residue field.

$$k((0)) = \mathbb{Q}.$$

Thus:

$$\mathbb{Z} \longrightarrow D(6) \longrightarrow \mathbb{Z}[1/6] \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}.$$

This is the arithmetic analogue of the generic point of $\operatorname{Spec}(k[x])$.

Example 6: $A = \mathbb{F}_p[t]$, $f = t$, $\mathfrak{p} = (t-a)$ with $a \neq 0$

Let

$$A = \mathbb{F}_p[t], \qquad f = t, \qquad \mathfrak{p} = (t-a), \qquad a \in \mathbb{F}_p^\times.$$

Since $a \neq 0$, we have

$$t \notin (t-a),$$

so

$$(t-a) \in D(t).$$

Global ring.

$$A = \mathbb{F}_p[t].$$

Distinguished open.

$$D(t) = \operatorname{Spec}(\mathbb{F}_p[t]) \setminus V(t).$$

Localization on the open.

$$A_t = \mathbb{F}_p[t, t^{-1}].$$

Local ring at the point.

$$A_{(t-a)} = \mathbb{F}_p[t]_{(t-a)}.$$

Residue field.

$$k((t-a)) \cong \mathbb{F}_p.$$

Thus:

$$\mathbb{F}_p[t] \longrightarrow D(t) \longrightarrow \mathbb{F}_p[t, t^{-1}] \longrightarrow \mathbb{F}_p[t]_{(t-a)} \longrightarrow \mathbb{F}_p.$$

Example 7: $A = \mathbb{F}_p$

Now consider a field itself.

Let

$$A = \mathbb{F}_p.$$

Since A is a field, the only prime ideal is

$$(0).$$

So
$$\operatorname{Spec}(\mathbb{F}_p)$$
has exactly one point.

Take any nonzero element
$$f \in \mathbb{F}_p^\times.$$
Then f is invertible, so
$$D(f) = \operatorname{Spec}(\mathbb{F}_p).$$

Also,
$$A_f = A, \qquad A_{(0)} = A, \qquad k((0)) = A = \mathbb{F}_p.$$

So the chain becomes
$$\mathbb{F}_p \longrightarrow \operatorname{Spec}(\mathbb{F}_p) \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{F}_p.$$

This is the simplest one-point affine scheme.

Example 8: $A = \mathbb{Z}/6\mathbb{Z}$

This is a finite reduced ring which is not a domain.

By the Chinese remainder theorem,
$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{F}_2 \times \mathbb{F}_3.$$

So $\operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$ has two prime ideals, corresponding to the two factors.

Take
$$A = \mathbb{Z}/6\mathbb{Z}, \qquad f = \overline{2}, \qquad \mathfrak{p} = (3)/(6).$$
Then $\overline{2} \notin \mathfrak{p}$, so
$$\mathfrak{p} \in D(\overline{2}).$$

Global ring.
$$A = \mathbb{Z}/6\mathbb{Z}.$$

Distinguished open.
$$D(\overline{2})$$
contains only the point corresponding to $(3)/(6)$.

Localization on the open. Localizing at $\overline{2}$ kills the $\mathbb{F}_2$-component and keeps the $\mathbb{F}_3$-component, so
$$A_{\overline{2}} \cong \mathbb{F}_3.$$

Local ring at the point.
$$A_{\mathfrak{p}} \cong \mathbb{F}_3.$$

Residue field.

$$k(\mathfrak{p}) \cong \mathbb{F}_3.$$

Thus:

$$\mathbb{Z}/6\mathbb{Z} \longrightarrow D(\overline{2}) \longrightarrow \mathbb{F}_3 \longrightarrow \mathbb{F}_3 \longrightarrow \mathbb{F}_3.$$

This example is useful because localization isolates one connected component of a product ring.

Example 9: $A = \mathbb{Z}/4\mathbb{Z}$

This is the finite nonreduced example.

Let

$$A = \mathbb{Z}/4\mathbb{Z}.$$

The only prime ideal is

$$(2)/(4).$$

So $\operatorname{Spec}(A)$ has exactly one point.

Take

$$f = \overline{2}.$$

Then

$$\overline{2}^2 = \overline{4} = 0,$$

so $\overline{2}$ is nilpotent. Every nilpotent element lies in every prime ideal, hence

$$D(\overline{2}) = \varnothing.$$

This is an important scheme-theoretic phenomenon: a nonzero element can define the empty distinguished open.

Now localizing at the unique prime gives

$$A_{(2)/(4)} = A,$$

because A is already local.

Its residue field is

$$k((2)/(4)) \cong \mathbb{F}_2.$$

So this example gives:

- one point topologically,
- a nonreduced local ring,
- an empty distinguished open from a nonzero nilpotent element.

Example 10: $A = \mathbb{Z}[x]$**,** $f = x$**,** $\mathfrak{p} = (2, x-1)$

This example mixes arithmetic and polynomial geometry.

Let

$$A = \mathbb{Z}[x], \qquad f = x, \qquad \mathfrak{p} = (2, x-1).$$

We check that

$$x \notin (2, x-1).$$

Indeed, modulo the ideal $(2, x-1)$, the class of x is equal to 1, not 0. Hence

$$\mathfrak{p} \in D(x).$$

Global ring.

$$A = \mathbb{Z}[x].$$

Distinguished open.

$$D(x) = \{\mathfrak{q} \in \operatorname{Spec}(\mathbb{Z}[x]) : x \notin \mathfrak{q}\}.$$

Localization on the open.

$$A_x = \mathbb{Z}[x, 1/x].$$

Local ring at the point.

$$A_{\mathfrak{p}} = \mathbb{Z}[x]_{(2,x-1)}.$$

Residue field.

$$k(\mathfrak{p}) \cong \mathbb{F}_2.$$

Thus:

$$\mathbb{Z}[x] \longrightarrow D(x) \longrightarrow \mathbb{Z}[x, 1/x] \longrightarrow \mathbb{Z}[x]_{(2,x-1)} \longrightarrow \mathbb{F}_2.$$

This is a useful mixed example because it combines polynomial coordinates and reduction modulo a prime.

Comparison table

For quick reference, here is a summary of the main examples:

Ring A	**Element** f	**Prime** $\mathfrak{p}$	**Local ring** $A_{\mathfrak{p}}$	**Residue field** $k(\mathfrak{p})$
$k[x]$	x	$(x-a)$, $a \neq 0$	$k[x]_{(x-a)}$	k
$k[x]$	x	(0)	$k(x)$	$k(x)$

Ring A	**Element** f	**Prime** $\mathfrak{p}$	**Local ring** $A_\mathfrak{p}$	**Residue field** $k(\mathfrak{p})$
$k[x,y]$	x	$(x-1,y)$	$k[x,y]_{(x-1,y)}$	k
$\mathbb{Z}$	2	(3)	$\mathbb{Z}_{(3)}$	$\mathbb{F}_3$
$\mathbb{Z}$	6	(0)	$\mathbb{Q}$	$\mathbb{Q}$
$\mathbb{F}_p[t]$	t	$(t-a)$, $a \neq 0$	$\mathbb{F}_p[t]_{(t-a)}$	$\mathbb{F}_p$
$\mathbb{F}_p$	$f \neq 0$	(0)	$\mathbb{F}_p$	$\mathbb{F}_p$
$\mathbb{Z}/6\mathbb{Z}$	$\overline{2}$	$(3)/(6)$	$\mathbb{F}_3$	$\mathbb{F}_3$
$\mathbb{Z}/4\mathbb{Z}$	$\overline{2}$	$(2)/(4)$	$\mathbb{Z}/4\mathbb{Z}$	$\mathbb{F}_2$
$\mathbb{Z}[x]$	x	$(2,x-1)$	$\mathbb{Z}[x]_{(2,x-1)}$	$\mathbb{F}_2$

Final summary

These examples show several different behaviors of the chain

$$A \longrightarrow D(f) \longrightarrow A_f \longrightarrow A_\mathfrak{p} \longrightarrow k(\mathfrak{p}).$$

In particular, they illustrate that:

- polynomial rings over fields behave like classical affine geometry,
- $\mathbb{Z}$ behaves like an arithmetic affine line,
- fields give one-point spectra,
- finite product rings such as $\mathbb{Z}/6\mathbb{Z}$ have disconnected spectra,
- nonreduced rings such as $\mathbb{Z}/4\mathbb{Z}$ can have nilpotents and empty distinguished opens coming from nonzero elements,
- mixed rings such as $\mathbb{Z}[x]$ combine arithmetic and geometric features.

Together these examples give a broad and concrete picture of how localization and local rings behave in algebraic geometry and arithmetic geometry.

2.19 A collection of step-by-step examples for exact sequences

Exact sequences are one of the main tools for understanding ideals, quotients, localization, tensor products, and local computations.

The guiding pattern is:

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

which is called a *short exact sequence*.

It means:

- the first map is injective,
- the last map is surjective,
- the image of the first map is exactly the kernel of the second map.

So M is built by placing the subobject M' inside M and then quotienting by it to obtain M''.

The following examples show how exact sequences arise in polynomial rings, in $\mathbb{Z}$, in quotient rings, under localization, and under tensoring.

Example 1: the basic quotient sequence

Let A be a ring and $I \subseteq A$ an ideal. Then there is always a short exact sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0.$$

Here:

- the map $I \to A$ is the inclusion,
- the map $A \to A/I$ is the quotient map $a \mapsto a + I$.

This is exact because:

- the inclusion $I \hookrightarrow A$ is injective,
- the quotient map $A \to A/I$ is surjective,
- the kernel of $A \to A/I$ is exactly I.

This is the most basic exact sequence attached to an ideal.

Example 2: $0 \to (x) \to k[x] \to k[x]/(x) \to 0$

Take

$$A = k[x], \qquad I = (x).$$

Then the basic quotient sequence becomes

$$0 \longrightarrow (x) \longrightarrow k[x] \longrightarrow k[x]/(x) \longrightarrow 0.$$

Now

$$k[x]/(x) \cong k,$$

so we get

$$0 \longrightarrow (x) \longrightarrow k[x] \longrightarrow k \longrightarrow 0.$$

The last map is evaluation at $x = 0$:

$$f(x) \longmapsto f(0).$$

Its kernel is the set of polynomials vanishing at 0, that is, the ideal (x). Hence the sequence is exact.

This is one of the most useful exact sequences in elementary algebraic geometry.

Example 3: multiplication by x on $k[x]$

Consider the sequence

$$0 \longrightarrow k[x] \xrightarrow{\cdot x} k[x] \longrightarrow k[x]/(x) \longrightarrow 0,$$

where the first map is multiplication by x.

This is exact.

Indeed:

- multiplication by x is injective because $k[x]$ is a domain,
- the image of multiplication by x is exactly (x),
- the quotient by (x) is $k[x]/(x) \cong k$.

So we also have the concrete short exact sequence

$$0 \longrightarrow k[x] \xrightarrow{\cdot x} k[x] \longrightarrow k \longrightarrow 0.$$

This is often more useful than the inclusion sequence because both first two terms are the same module.

Example 4: multiplication by n on $\mathbb{Z}$

For any integer $n \geq 1$, there is a short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

Indeed:

- multiplication by n is injective on $\mathbb{Z}$,
- its image is the ideal $(n) \subseteq \mathbb{Z}$,
- the quotient is $\mathbb{Z}/(n) \cong \mathbb{Z}/n\mathbb{Z}$.

So exactness expresses the elementary fact that

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}.$$

This is the arithmetic analogue of the polynomial example above.

Example 5: $0 \to 2\mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$

The special case $n = 2$ gives

$$0 \longrightarrow 2\mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Equivalently, via multiplication by 2,

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

The quotient map sends an integer to its parity class mod 2. Its kernel consists of the even integers, which is exactly $2\mathbb{Z}$.

Example 6: the sequence for a maximal ideal in $k[x, y]$

Take

$$A = k[x, y], \qquad \mathfrak{m} = (x - a, y - b).$$

Then

$$0 \longrightarrow \mathfrak{m} \longrightarrow k[x, y] \longrightarrow k[x, y]/\mathfrak{m} \longrightarrow 0$$

is exact.

Since

$$k[x, y]/(x - a, y - b) \cong k,$$

this becomes

$$0 \longrightarrow (x - a, y - b) \longrightarrow k[x, y] \longrightarrow k \longrightarrow 0.$$

The last map is evaluation at the point (a, b):

$$f(x, y) \longmapsto f(a, b).$$

Its kernel is exactly the ideal of polynomials vanishing at (a, b).

Example 7: a nonreduced quotient $k[x]/(x^2)$

Take

$$A = k[x], \qquad I = (x^2).$$

Then

$$0 \longrightarrow (x^2) \longrightarrow k[x] \longrightarrow k[x]/(x^2) \longrightarrow 0$$

is exact.

This quotient is not reduced, because the class of x satisfies

$$\overline{x} \neq 0, \qquad \overline{x}^2 = 0.$$

So exact sequences also detect nonreduced quotients naturally.

Example 8: the exact sequence $0 \to (x) \to k[x]/(x^2) \to k \to 0$

Inside the ring

$$A = k[x]/(x^2),$$

consider the ideal generated by $\overline{x}$. Then there is a short exact sequence

$$0 \longrightarrow (\overline{x}) \longrightarrow k[x]/(x^2) \longrightarrow k \longrightarrow 0.$$

Indeed, the quotient map sends the class of a polynomial to its constant term, and its kernel is the ideal generated by $\overline{x}$.

Now $(\overline{x}) \cong k$ as a vector space over k, but not as a subring of A. So this exact sequence shows that the doubled point is built from a reduced point plus a nilpotent infinitesimal part.

Example 9: an exact sequence that is *not* split

The sequence

$$0 \longrightarrow (\overline{x}) \longrightarrow k[x]/(x^2) \longrightarrow k \longrightarrow 0$$

is exact, but it does not split as a sequence of rings.

Indeed,

$$k[x]/(x^2) \not\cong k \times k$$

as rings, because $k[x]/(x^2)$ has a nonzero nilpotent element, whereas $k \times k$ has no nonzero nilpotents.

So exact does not automatically mean split.

This is an important counterexample.

Example 10: a sequence that fails to be exact

Consider

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}.$$

This sequence is *not* exact at the middle copy of $\mathbb{Z}$.

Indeed:

- the image of the first map is $2\mathbb{Z}$,
- the kernel of the second map is

  $$\ker(\cdot 2) = \{0\},$$

  because $\mathbb{Z}$ has no nonzero 2-torsion.

So

$$\operatorname{im}(\cdot 2) = 2\mathbb{Z} \neq \{0\} = \ker(\cdot 2).$$

Thus the sequence is not exact.

This is a good reminder that composing two maps does not automatically give exactness.

Example 11: exactness after localization

Exact sequences behave very well under localization.

Starting from

$$0 \longrightarrow (x) \longrightarrow k[x] \longrightarrow k \longrightarrow 0,$$

localize at the multiplicative set generated by x. Then one gets

$$0 \longrightarrow (x)_x \longrightarrow k[x]_x \longrightarrow k_x \longrightarrow 0.$$

Now

$$k[x]_x = k[x, x^{-1}],$$

and since x acts as zero on k, localizing k at x kills it:

$$k_x = 0.$$

Also $(x)_x = k[x]_x$, because x becomes invertible in $k[x]_x$. So the sequence becomes

$$0 \longrightarrow k[x]_x \xrightarrow{\cong} k[x]_x \longrightarrow 0 \longrightarrow 0.$$

This is exact.

Thus localization preserves exact sequences, but the terms themselves may change drastically.

Example 12: exactness after localization at a prime

Take again

$$0 \longrightarrow (x) \longrightarrow k[x] \longrightarrow k \longrightarrow 0,$$

and localize at the prime ideal (x). Then we get

$$0 \longrightarrow (x)_{(x)} \longrightarrow k[x]_{(x)} \longrightarrow k_{(x)} \longrightarrow 0.$$

Since k is the quotient $k[x]/(x)$, localizing at (x) does not change it, so

$$k_{(x)} \cong k.$$

Hence

$$0 \longrightarrow (x)k[x]_{(x)} \longrightarrow k[x]_{(x)} \longrightarrow k \longrightarrow 0$$

is exact.

This is the local version of the exact sequence at the closed point $x = 0$.

Example 13: localization of $\mathbb{Z}$-sequences

Start from

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 6} \mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z} \longrightarrow 0.$$

Localize at the prime (5). Then

$$0 \longrightarrow \mathbb{Z}_{(5)} \xrightarrow{\cdot 6} \mathbb{Z}_{(5)} \longrightarrow (\mathbb{Z}/6\mathbb{Z})_{(5)} \longrightarrow 0$$

is exact.

Now 6 is invertible in $\mathbb{Z}_{(5)}$, so multiplication by 6 is an isomorphism. Hence

$$(\mathbb{Z}/6\mathbb{Z})_{(5)} = 0.$$

So at the prime (5), the quotient $\mathbb{Z}/6\mathbb{Z}$ disappears.

By contrast, localizing at (2) or (3) gives a nonzero module.

This is a very useful arithmetic example showing that localization isolates behavior at one prime.

Example 14: exact sequences and tensoring

Tensor products do *not* preserve exactness on the left in general, but they do preserve right exactness.

Start from the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Now tensor with $\mathbb{Z}/2\mathbb{Z}$. One gets

$$\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \longrightarrow (\mathbb{Z}/2\mathbb{Z}) \otimes \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Since

$$\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z},$$

this becomes

$$\mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

The left map is zero, so the sequence is

$$\mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

This is not exact on the left, because the kernel of the zero map is all of $\mathbb{Z}/2\mathbb{Z}$, while the image of the previous zero term would be 0.

So tensoring destroyed left exactness.

This is the standard counterexample showing that tensor products are right exact but not always exact.

Example 15: tensoring a quotient sequence over a field

Now take a field k, and consider

$$0 \longrightarrow (x) \longrightarrow k[x] \longrightarrow k \longrightarrow 0.$$

Tensor with a field extension K/k. Since tensoring over a field is exact, we get

$$0 \longrightarrow (x) \otimes_k K \longrightarrow k[x] \otimes_k K \longrightarrow k \otimes_k K \longrightarrow 0.$$

Using

$$k[x] \otimes_k K \cong K[x], \qquad k \otimes_k K \cong K,$$

this becomes

$$0 \longrightarrow (x)K[x] \longrightarrow K[x] \longrightarrow K \longrightarrow 0.$$

So over a field, extension of scalars preserves short exact sequences of vector spaces and free modules much better than tensoring over $\mathbb{Z}$.

Example 16: exact sequence for a principal ideal in $k[x, y]$

Take
$$A = k[x, y], \qquad f = xy.$$
Then there is an exact sequence
$$0 \longrightarrow (xy) \longrightarrow k[x, y] \longrightarrow k[x, y]/(xy) \longrightarrow 0.$$

This quotient ring describes two crossing lines. So the exact sequence packages the passage from the ambient affine plane to the closed subscheme cut out by $xy = 0$.

Example 17: a useful kernel-image sequence

Let
$$\varphi : k[x, y] \longrightarrow k[x]$$
be the map given by
$$\varphi(f(x, y)) = f(x, 0).$$

Then φ is surjective, and its kernel is (y). Hence
$$0 \longrightarrow (y) \longrightarrow k[x, y] \xrightarrow{\varphi} k[x] \longrightarrow 0$$
is exact.

Geometrically, this is restriction from the affine plane to the x-axis.

Similarly, evaluation at $x = a$ gives
$$0 \longrightarrow (x - a) \longrightarrow k[x] \longrightarrow k \longrightarrow 0.$$

Example 18: exact sequence for the residue field at a local ring

Let
$$A_{\mathfrak{p}}$$
be a local ring with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. Then we always have
$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow k(\mathfrak{p}) \longrightarrow 0.$$

This is the local version of the basic quotient sequence.

For example, in
$$k[x]_{(x-a)},$$
we get
$$0 \longrightarrow (x - a)k[x]_{(x-a)} \longrightarrow k[x]_{(x-a)} \longrightarrow k \longrightarrow 0.$$

Thus the residue field is the quotient of the local ring by the ideal of functions vanishing at the point.

Comparison table

For quick reference, here are the main exact sequences above:

Exact sequence	Interpretation
$0 \to I \to A \to A/I \to 0$	Basic quotient sequence
$0 \to (x) \to k[x] \to k \to 0$	Evaluation at $x = 0$
$0 \to k[x] \xrightarrow{\cdot x} k[x] \to k \to 0$	Multiplication by x
$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$	Arithmetic quotient sequence
$0 \to (x-a, y-b) \to k[x,y] \to k \to 0$	Evaluation at (a, b)
$0 \to (x^2) \to k[x] \to k[x]/(x^2) \to 0$	Nonreduced quotient
$0 \to (\bar{x}) \to k[x]/(x^2) \to k \to 0$	Doubled point mod its nilpotent part
$0 \to (y) \to k[x,y] \to k[x] \to 0$	Restriction to the x-axis
$0 \to \mathfrak{p}A_\mathfrak{p} \to A_\mathfrak{p} \to k(\mathfrak{p}) \to 0$	Residue field from local ring

Final summary

These examples show several key principles.

- Exact sequences are the natural language for subobjects and quotients.
- Ideals and quotient rings always produce short exact sequences.
- Localization preserves exactness.
- Tensor products preserve right exactness, but may destroy left exactness.
- Local rings fit into exact sequences whose quotient is the residue field.
- Nonreduced schemes and reducible schemes appear naturally inside exact sequences.

Thus exact sequences connect directly with the whole scheme-theoretic picture:

$$\text{ideals} \longrightarrow \text{quotients} \longrightarrow \text{local rings} \longrightarrow \text{residue fields} \longrightarrow \text{localization and tensoring.}$$

They are one of the central organizing tools of algebra, algebraic geometry, and commutative algebra.

2.20 Concrete exact sequences over rings such as $\mathbb{Z}$, $\mathbb{Z}/p\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}_{(p)}$, $\mathbb{Z}_p$, and $\mathbb{F}_p[x]$

It is very useful to rewrite the abstract exact sequences in completely concrete rings. This makes the meaning of exactness much easier to see.

Recall that a short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

means that M' injects into M, and that the quotient of M by the image of M' is M''.

Below are several concrete examples.

Example 1: the basic quotient sequence over $\mathbb{Z}$

For any integer $n \geq 1$, there is a short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

This is exact because:

- the map $m \mapsto nm$ is injective,
- its image is $n\mathbb{Z}$,
- the quotient of $\mathbb{Z}$ by $n\mathbb{Z}$ is $\mathbb{Z}/n\mathbb{Z}$.

Equivalently, one may write

$$0 \longrightarrow (n) \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/(n) \longrightarrow 0.$$

Example 2: the case $n = p$ prime

If p is prime, then

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot p} \mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow 0$$

is the standard exact sequence whose quotient is the field $\mathbb{F}_p$.

This is the arithmetic analogue of

$$0 \longrightarrow (x - a) \longrightarrow k[x] \longrightarrow k \longrightarrow 0.$$

Example 3: the field $\mathbb{Z}/p\mathbb{Z}$

Let

$$A = \mathbb{Z}/p\mathbb{Z}.$$

Since A is a field, its only prime ideal is (0), and its residue field is itself.

So the natural exact sequence is

$$0 \longrightarrow 0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow 0.$$

This says that the whole ring is already a field, so there is no nontrivial maximal ideal to quotient by.

Also, if $\bar{a} \in \mathbb{Z}/p\mathbb{Z}$ is nonzero, then multiplication by $\bar{a}$ is an isomorphism, so

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \xrightarrow{\cdot \bar{a}} \mathbb{Z}/p\mathbb{Z} \longrightarrow 0$$

is exact.

Example 4: the nonreduced ring $\mathbb{Z}/4\mathbb{Z}$

Let
$$A = \mathbb{Z}/4\mathbb{Z}.$$
Then the class $\bar{2}$ is nonzero, but
$$\bar{2}^2 = \bar{4} = \bar{0}.$$
So $\bar{2}$ is nilpotent.

The ideal generated by $\bar{2}$ is
$$(\bar{2}) = \{\,\bar{0}, \bar{2}\,\}.$$
Hence there is a short exact sequence
$$0 \longrightarrow (\bar{2}) \longrightarrow \mathbb{Z}/4\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$
This is the finite-ring analogue of
$$0 \longrightarrow (\bar{x}) \longrightarrow k[x]/(x^2) \longrightarrow k \longrightarrow 0.$$
It expresses the fact that $\mathbb{Z}/4\mathbb{Z}$ is a one-point nonreduced local ring whose residue field is $\mathbb{F}_2$.

Example 5: multiplication by $\bar{2}$ on $\mathbb{Z}/4\mathbb{Z}$

There is also an exact sequence
$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow \mathbb{Z}/4\mathbb{Z} \xrightarrow{\cdot\bar{2}} \mathbb{Z}/4\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$
Indeed:

- the kernel of multiplication by $\bar{2}$ is $\{\,\bar{0}, \bar{2}\,\} \cong \mathbb{Z}/2\mathbb{Z}$,
- the image is also $\{\,\bar{0}, \bar{2}\,\}$,
- the cokernel is again $\mathbb{Z}/2\mathbb{Z}$.

This gives a very concrete finite exact sequence where torsion and nilpotents are visible.

Example 6: the reduced but disconnected ring $\mathbb{Z}/6\mathbb{Z}$

Let
$$A = \mathbb{Z}/6\mathbb{Z}.$$
By the Chinese remainder theorem,
$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$
So this ring is reduced, but not a domain. It has two components.

Now the ideal $(\bar{2})$ has quotient
$$(\mathbb{Z}/6\mathbb{Z})/(\bar{2}) \cong \mathbb{Z}/2\mathbb{Z},$$

so we obtain

$$0 \longrightarrow (\bar{2}) \longrightarrow \mathbb{Z}/6\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Similarly,

$$0 \longrightarrow (\bar{3}) \longrightarrow \mathbb{Z}/6\mathbb{Z} \longrightarrow \mathbb{Z}/3\mathbb{Z} \longrightarrow 0.$$

These two sequences isolate the two different residue-field pieces of $\mathbb{Z}/6\mathbb{Z}$.

Example 7: the ring $\mathbb{Z}/8\mathbb{Z}$

Let

$$A = \mathbb{Z}/8\mathbb{Z}.$$

Then $(\bar{2})$ is a proper ideal, and the quotient is

$$(\mathbb{Z}/8\mathbb{Z})/(\bar{2}) \cong \mathbb{Z}/2\mathbb{Z}.$$

So:

$$0 \longrightarrow (\bar{2}) \longrightarrow \mathbb{Z}/8\mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0.$$

Likewise,

$$(\mathbb{Z}/8\mathbb{Z})/(\bar{4}) \cong \mathbb{Z}/4\mathbb{Z},$$

so:

$$0 \longrightarrow (\bar{4}) \longrightarrow \mathbb{Z}/8\mathbb{Z} \longrightarrow \mathbb{Z}/4\mathbb{Z} \longrightarrow 0.$$

These are useful examples of exact sequences in rings with higher nilpotent structure.

Example 8: the local ring $\mathbb{Z}_{(p)}$

Let

$$A = \mathbb{Z}_{(p)}.$$

This is the localization of $\mathbb{Z}$ at the prime ideal (p).

Its unique maximal ideal is

$$p\mathbb{Z}_{(p)}.$$

Its residue field is

$$\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{F}_p.$$

So the fundamental local exact sequence is

$$0 \longrightarrow p\mathbb{Z}_{(p)} \longrightarrow \mathbb{Z}_{(p)} \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is the arithmetic model of the local sequence

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow k(\mathfrak{p}) \longrightarrow 0.$$

Example 9: the p-adic integers $\mathbb{Z}_p$

Now let

$$A = \mathbb{Z}_p,$$

the ring of p-adic integers.

Again A is a local ring with maximal ideal

$$p\mathbb{Z}_p$$

and residue field

$$\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{F}_p.$$

So we get the short exact sequence

$$0 \longrightarrow p\mathbb{Z}_p \longrightarrow \mathbb{Z}_p \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

More generally, for every $n \geq 1$,

$$0 \longrightarrow p^n\mathbb{Z}_p \longrightarrow \mathbb{Z}_p \longrightarrow \mathbb{Z}/p^n\mathbb{Z} \longrightarrow 0.$$

These are among the most important exact sequences in p-adic algebra and arithmetic geometry.

Example 10: polynomial ring over a finite field

Let

$$A = \mathbb{F}_p[x].$$

If $a \in \mathbb{F}_p$, then evaluation at $x = a$ gives a surjective ring homomorphism

$$\mathbb{F}_p[x] \longrightarrow \mathbb{F}_p, \qquad f(x) \longmapsto f(a),$$

whose kernel is $(x - a)$. Hence we have the exact sequence

$$0 \longrightarrow (x - a) \longrightarrow \mathbb{F}_p[x] \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is exactly analogous to

$$0 \longrightarrow (x - a) \longrightarrow k[x] \longrightarrow k \longrightarrow 0$$

over an arbitrary field k.

Example 11: irreducible polynomial over $\mathbb{F}_p[x]$

Let $f(x) \in \mathbb{F}_p[x]$ be irreducible of degree d. Then

$$0 \longrightarrow (f) \longrightarrow \mathbb{F}_p[x] \longrightarrow \mathbb{F}_p[x]/(f) \longrightarrow 0$$

is exact.

Since f is irreducible, the quotient is a field:

$$\mathbb{F}_p[x]/(f) \cong \mathbb{F}_{p^d}.$$

So in degree 2, for example, one gets

$$0 \longrightarrow (f) \longrightarrow \mathbb{F}_p[x] \longrightarrow \mathbb{F}_{p^2} \longrightarrow 0.$$

This is the finite-field analogue of

$$0 \longrightarrow (x^2+1) \longrightarrow \mathbb{R}[x] \longrightarrow \mathbb{C} \longrightarrow 0.$$

Example 12: mixed arithmetic-polynomial ring $\mathbb{Z}[x]$

Let

$$A = \mathbb{Z}[x].$$

Take the maximal ideal

$$\mathfrak{m} = (p, x-a),$$

where p is prime and $a \in \mathbb{Z}$.

Then the quotient map

$$\mathbb{Z}[x] \longrightarrow \mathbb{F}_p, \qquad f(x) \longmapsto f(a) \bmod p$$

has kernel $(p, x-a)$. Therefore there is a short exact sequence

$$0 \longrightarrow (p, x-a) \longrightarrow \mathbb{Z}[x] \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is a very concrete example combining arithmetic reduction modulo p and geometric evaluation at $x = a$.

Comparison table

For quick reference, here is a table parallel to the earlier one, but now using concrete rings.

Exact sequence	**Interpretation**
$0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$	Basic quotient over $\mathbb{Z}$
$0 \to \mathbb{Z} \xrightarrow{\cdot p} \mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$	Prime quotient over $\mathbb{Z}$
$0 \to 0 \to \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z} \to 0$	A field is already its own residue field
$0 \to (\bar{2}) \to \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$	Finite nonreduced quotient
$0 \to \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z} \xrightarrow{\cdot \bar{2}} \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$	Kernel-image-cokernel sequence in $\mathbb{Z}/4\mathbb{Z}$

Exact sequence	**Interpretation**
$0 \to (\bar{2}) \to \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$	One component of $\mathbb{Z}/6\mathbb{Z}$
$0 \to (\bar{3}) \to \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \to 0$	Other component of $\mathbb{Z}/6\mathbb{Z}$
$0 \to p\mathbb{Z}_{(p)} \to \mathbb{Z}_{(p)} \to \mathbb{F}_p \to 0$	Residue field from the local ring $\mathbb{Z}_{(p)}$
$0 \to p\mathbb{Z}_p \to \mathbb{Z}_p \to \mathbb{F}_p \to 0$	Residue field from the p-adic integers
$0 \to p^n\mathbb{Z}_p \to \mathbb{Z}_p \to \mathbb{Z}/p^n\mathbb{Z} \to 0$	Higher p-power quotient in $\mathbb{Z}_p$
$0 \to (x-a) \to \mathbb{F}_p[x] \to \mathbb{F}_p \to 0$	Evaluation at $x = a$ over $\mathbb{F}_p$
$0 \to (f) \to \mathbb{F}_p[x] \to \mathbb{F}_{p^d} \to 0$	Quotient by an irreducible polynomial of degree d
$0 \to (p, x-a) \to \mathbb{Z}[x] \to \mathbb{F}_p \to 0$	Mixed arithmetic-geometric evaluation

Final summary

These concrete exact sequences show several different phenomena:

- over $\mathbb{Z}$, exactness expresses divisibility and quotient by n,
- over $\mathbb{Z}/p\mathbb{Z}$, the ring is already a field,
- over $\mathbb{Z}/4\mathbb{Z}$, nilpotents appear,
- over $\mathbb{Z}/6\mathbb{Z}$, one sees disconnected reduced components,
- over $\mathbb{Z}_{(p)}$ and $\mathbb{Z}_p$, one sees local rings and residue fields,
- over $\mathbb{F}_p[x]$, one sees points and finite field extensions,
- over $\mathbb{Z}[x]$, one sees arithmetic and geometry together.

So these exact sequences are concrete models for the abstract pattern

$$0 \longrightarrow \text{ideal} \longrightarrow \text{ring} \longrightarrow \text{quotient} \longrightarrow 0,$$

and they connect directly with local rings, residue fields, finite rings, polynomial rings, and arithmetic geometry.

2.21 Concrete local exact sequences of the form $0 \to \mathfrak{p}A_\mathfrak{p} \to A_\mathfrak{p} \to k(\mathfrak{p}) \to$

For every prime ideal $\mathfrak{p} \in \mathrm{Spec}(A)$, the local ring $A_\mathfrak{p}$ has maximal ideal

$$\mathfrak{p}A_\mathfrak{p},$$

and its residue field is

$$k(\mathfrak{p}) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}.$$

Therefore one always has a short exact sequence

$$0 \longrightarrow \mathfrak{p}A_\mathfrak{p} \longrightarrow A_\mathfrak{p} \longrightarrow k(\mathfrak{p}) \longrightarrow 0.$$

This sequence expresses the idea that the residue field is obtained by quotienting the local ring by the ideal of functions vanishing at the point.

Below are concrete versions of this sequence for several standard rings.

Example 1: $A = \mathbb{Z}$, $\mathfrak{p} = (p)$

Let

$$A = \mathbb{Z}, \qquad \mathfrak{p} = (p),$$

where p is a prime number.

Then the local ring is

$$A_{\mathfrak{p}} = \mathbb{Z}_{(p)}.$$

Its unique maximal ideal is

$$\mathfrak{p}A_{\mathfrak{p}} = p\mathbb{Z}_{(p)}.$$

Its residue field is

$$k(\mathfrak{p}) = \mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{F}_p.$$

So the exact sequence is

$$0 \longrightarrow p\mathbb{Z}_{(p)} \longrightarrow \mathbb{Z}_{(p)} \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is the standard local arithmetic example.

Example 2: $A = \mathbb{Z}$, $\mathfrak{p} = (0)$

Now take the generic point of $\mathrm{Spec}(\mathbb{Z})$:

$$\mathfrak{p} = (0).$$

Since $\mathbb{Z}$ is a domain,

$$A_{\mathfrak{p}} = \mathbb{Z}_{(0)} = \mathbb{Q}.$$

The maximal ideal is

$$(0)\mathbb{Q} = 0,$$

because $\mathbb{Q}$ is already a field.

So the residue field is

$$k((0)) = \mathbb{Q}.$$

Thus the exact sequence becomes

$$0 \longrightarrow 0 \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q} \longrightarrow 0.$$

This is the generic-point version, where the local ring is already a field.

Example 3: $A = \mathbb{Z}_p$

Let

$$A = \mathbb{Z}_p,$$

the ring of p-adic integers.

This is a local ring whose unique maximal ideal is

$$p\mathbb{Z}_p.$$

Its residue field is

$$\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{F}_p.$$

So the exact sequence is

$$0 \longrightarrow p\mathbb{Z}_p \longrightarrow \mathbb{Z}_p \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is formally similar to the sequence for $\mathbb{Z}_{(p)}$, but now the ring is complete in the p-adic topology.

Example 4: $A = \mathbb{F}_p[x]$**,** $\mathfrak{p} = (x - a)$

Let

$$A = \mathbb{F}_p[x], \qquad \mathfrak{p} = (x - a),$$

where $a \in \mathbb{F}_p$.

Then the local ring is

$$A_{\mathfrak{p}} = \mathbb{F}_p[x]_{(x-a)}.$$

Its maximal ideal is

$$(x - a)\mathbb{F}_p[x]_{(x-a)}.$$

Its residue field is

$$k(\mathfrak{p}) \cong \mathbb{F}_p.$$

So the exact sequence is

$$0 \longrightarrow (x - a)\mathbb{F}_p[x]_{(x-a)} \longrightarrow \mathbb{F}_p[x]_{(x-a)} \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is the finite-field analogue of the local sequence at a rational point of the affine line.

Example 5: $A = \mathbb{F}_p[x]$**,** $\mathfrak{p} = (f)$ **with** f **irreducible**

Let

$$A = \mathbb{F}_p[x], \qquad \mathfrak{p} = (f),$$

where $f(x) \in \mathbb{F}_p[x]$ is irreducible of degree d.

Then

$$A_{\mathfrak{p}} = \mathbb{F}_p[x]_{(f)}, \qquad \mathfrak{p}A_{\mathfrak{p}} = f\mathbb{F}_p[x]_{(f)}.$$

The residue field is

$$k(\mathfrak{p}) \cong \mathbb{F}_p[x]/(f) \cong \mathbb{F}_{p^d}.$$

So the exact sequence is

$$0 \longrightarrow f\mathbb{F}_p[x]_{(f)} \longrightarrow \mathbb{F}_p[x]_{(f)} \longrightarrow \mathbb{F}_{p^d} \longrightarrow 0.$$

This is the finite-field-extension version of the local residue field sequence.

Example 6: $A = \mathbb{Z}[x]$**,** $\mathfrak{p} = (p, x-a)$

Let

$$A = \mathbb{Z}[x], \qquad \mathfrak{p} = (p, x-a),$$

where p is prime and $a \in \mathbb{Z}$.

Then the local ring is

$$A_{\mathfrak{p}} = \mathbb{Z}[x]_{(p,x-a)}.$$

Its maximal ideal is

$$(p, x-a)\mathbb{Z}[x]_{(p,x-a)}.$$

Its residue field is

$$k(\mathfrak{p}) \cong \mathbb{F}_p.$$

So the exact sequence is

$$0 \longrightarrow (p, x-a)\mathbb{Z}[x]_{(p,x-a)} \longrightarrow \mathbb{Z}[x]_{(p,x-a)} \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is a mixed arithmetic-geometric example.

Example 7: $A = \mathbb{Z}/4\mathbb{Z}$**,** $\mathfrak{p} = (\bar{2})$

Let

$$A = \mathbb{Z}/4\mathbb{Z}.$$

This is already a local ring, whose unique prime ideal is

$$\mathfrak{p} = (\bar{2}).$$

So

$$A_{\mathfrak{p}} = A = \mathbb{Z}/4\mathbb{Z}.$$

Its maximal ideal is

$$(\bar{2}),$$

and its residue field is

$$k(\mathfrak{p}) = (\mathbb{Z}/4\mathbb{Z})/(\bar{2}) \cong \mathbb{F}_2.$$

Thus the exact sequence is

$$0 \longrightarrow (\bar{2}) \longrightarrow \mathbb{Z}/4\mathbb{Z} \longrightarrow \mathbb{F}_2 \longrightarrow 0.$$

This is a concrete finite nonreduced local example.

Example 8: $A = \mathbb{Z}/6\mathbb{Z}$, $\mathfrak{p} = (\bar{2})$

Let

$$A = \mathbb{Z}/6\mathbb{Z}.$$

This ring is not local, so we localize at the prime ideal

$$\mathfrak{p} = (\bar{2}).$$

After localization one gets

$$A_{\mathfrak{p}} \cong \mathbb{F}_2.$$

Since $\mathbb{F}_2$ is already a field, its maximal ideal is

$$0,$$

and its residue field is

$$k(\mathfrak{p}) \cong \mathbb{F}_2.$$

So the exact sequence is

$$0 \longrightarrow 0 \longrightarrow \mathbb{F}_2 \longrightarrow \mathbb{F}_2 \longrightarrow 0.$$

Thus localizing at this prime isolates one field component.

Example 9: $A = \mathbb{Z}/6\mathbb{Z}$, $\mathfrak{p} = (\bar{3})$

Similarly, localizing at

$$\mathfrak{p} = (\bar{3})$$

gives

$$A_{\mathfrak{p}} \cong \mathbb{F}_3.$$

So the maximal ideal is 0, and the residue field is $\mathbb{F}_3$. Hence the exact sequence is

$$0 \longrightarrow 0 \longrightarrow \mathbb{F}_3 \longrightarrow \mathbb{F}_3 \longrightarrow 0.$$

So in $\mathbb{Z}/6\mathbb{Z}$, each prime sees only one field factor after localization.

Example 10: $A = \mathbb{Z}/p\mathbb{Z}$

Let
$$A = \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p.$$

Since this is already a field, its only prime ideal is
$$(0).$$

So
$$A_{(0)} = A = \mathbb{F}_p, \qquad (0)A_{(0)} = 0, \qquad k((0)) = \mathbb{F}_p.$$

Thus the exact sequence is
$$0 \longrightarrow 0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{F}_p \longrightarrow 0.$$

This is the simplest one-point field case.

Comparison table

For quick reference, here is a table summarizing the examples above.

Ring A	**Prime** $\mathfrak{p}$	**Local ring** $A_\mathfrak{p}$	**Maximal ideal** $\mathfrak{p}A_\mathfrak{p}$	**Residue field** $k(\mathfrak{p})$
$\mathbb{Z}$	(p)	$\mathbb{Z}_{(p)}$	$p\mathbb{Z}_{(p)}$	$\mathbb{F}_p$
$\mathbb{Z}$	(0)	$\mathbb{Q}$	0	$\mathbb{Q}$
$\mathbb{Z}_p$	$p\mathbb{Z}_p$	$\mathbb{Z}_p$	$p\mathbb{Z}_p$	$\mathbb{F}_p$
$\mathbb{F}_p[x]$	$(x-a)$	$\mathbb{F}_p[x]_{(x-a)}$	$(x-a)\mathbb{F}_p[x]_{(x-a)}$	$\mathbb{F}_p$
$\mathbb{F}_p[x]$	(f)	$\mathbb{F}_p[x]_{(f)}$	$f\mathbb{F}_p[x]_{(f)}$	$\mathbb{F}_{p^d}$
$\mathbb{Z}[x]$	$(p, x-a)$	$\mathbb{Z}[x]_{(p,x-a)}$	$(p, x-a)\mathbb{Z}[x]_{(p,x-a)}$	$\mathbb{F}_p$
$\mathbb{Z}/4\mathbb{Z}$	$(\bar{2})$	$\mathbb{Z}/4\mathbb{Z}$	$(\bar{2})$	$\mathbb{F}_2$
$\mathbb{Z}/6\mathbb{Z}$	$(\bar{2})$	$\mathbb{F}_2$	0	$\mathbb{F}_2$
$\mathbb{Z}/6\mathbb{Z}$	$(\bar{3})$	$\mathbb{F}_3$	0	$\mathbb{F}_3$
$\mathbb{Z}/p\mathbb{Z}$	(0)	$\mathbb{F}_p$	0	$\mathbb{F}_p$

Final summary

These examples illustrate two main cases.

Case 1: the local ring is not yet a field. Then the sequence has the form

$$0 \longrightarrow \text{maximal ideal} \longrightarrow \text{local ring} \longrightarrow \text{smaller residue field} \longrightarrow 0.$$

Typical examples are:

$$0 \longrightarrow p\mathbb{Z}_{(p)} \longrightarrow \mathbb{Z}_{(p)} \longrightarrow \mathbb{F}_p \longrightarrow 0,$$
$$0 \longrightarrow p\mathbb{Z}_p \longrightarrow \mathbb{Z}_p \longrightarrow \mathbb{F}_p \longrightarrow 0,$$
$$0 \longrightarrow (\bar{2}) \longrightarrow \mathbb{Z}/4\mathbb{Z} \longrightarrow \mathbb{F}_2 \longrightarrow 0.$$

Case 2: the local ring is already a field. Then the maximal ideal is 0, and the sequence degenerates to

$$0 \longrightarrow 0 \longrightarrow A_\mathfrak{p} \longrightarrow A_\mathfrak{p} \longrightarrow 0.$$

Typical examples are:

$$0 \longrightarrow 0 \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q} \longrightarrow 0,$$
$$0 \longrightarrow 0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{F}_p \longrightarrow 0,$$
$$0 \longrightarrow 0 \longrightarrow \mathbb{F}_2 \longrightarrow \mathbb{F}_2 \longrightarrow 0.$$

So the sequence

$$0 \longrightarrow \mathfrak{p}A_\mathfrak{p} \longrightarrow A_\mathfrak{p} \longrightarrow k(\mathfrak{p}) \longrightarrow 0$$

is the universal local exact sequence: it measures the passage from the local ring to the residue field by killing the functions that vanish at the point.